

Digital Logic Design

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Credit Hour: 3

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counter

Counters are used to count the clock pulses. The clock pulses occur at regular intervals. They are constructed with flip flops and logic gates.

They are of two types -

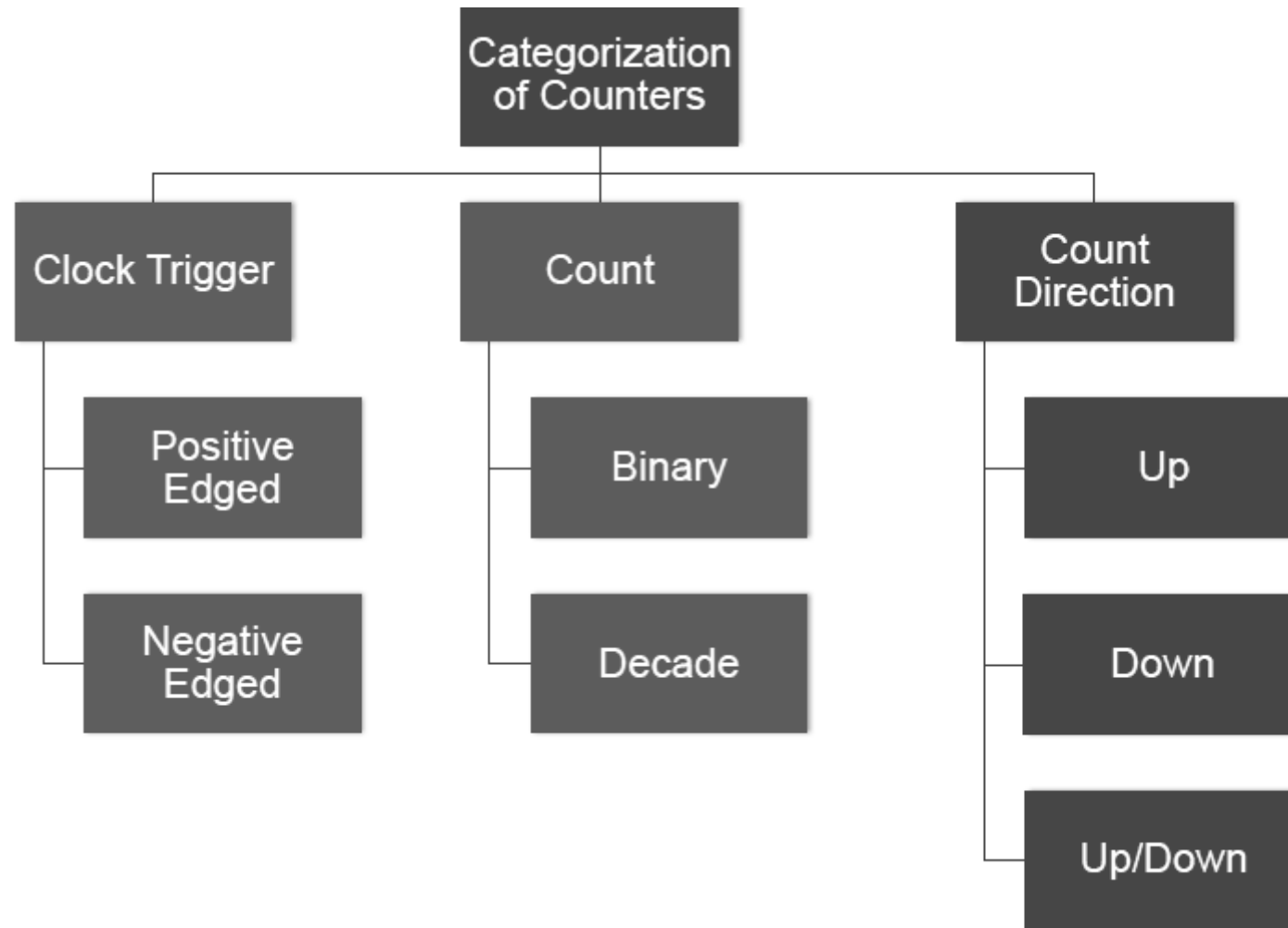
1. Asynchronous counter

In the asynchronous counter, an external clock pulse is provided for only the first flip flop, thereafter the output of the 1st FF acts as a clock pulse for the second FF and so on.

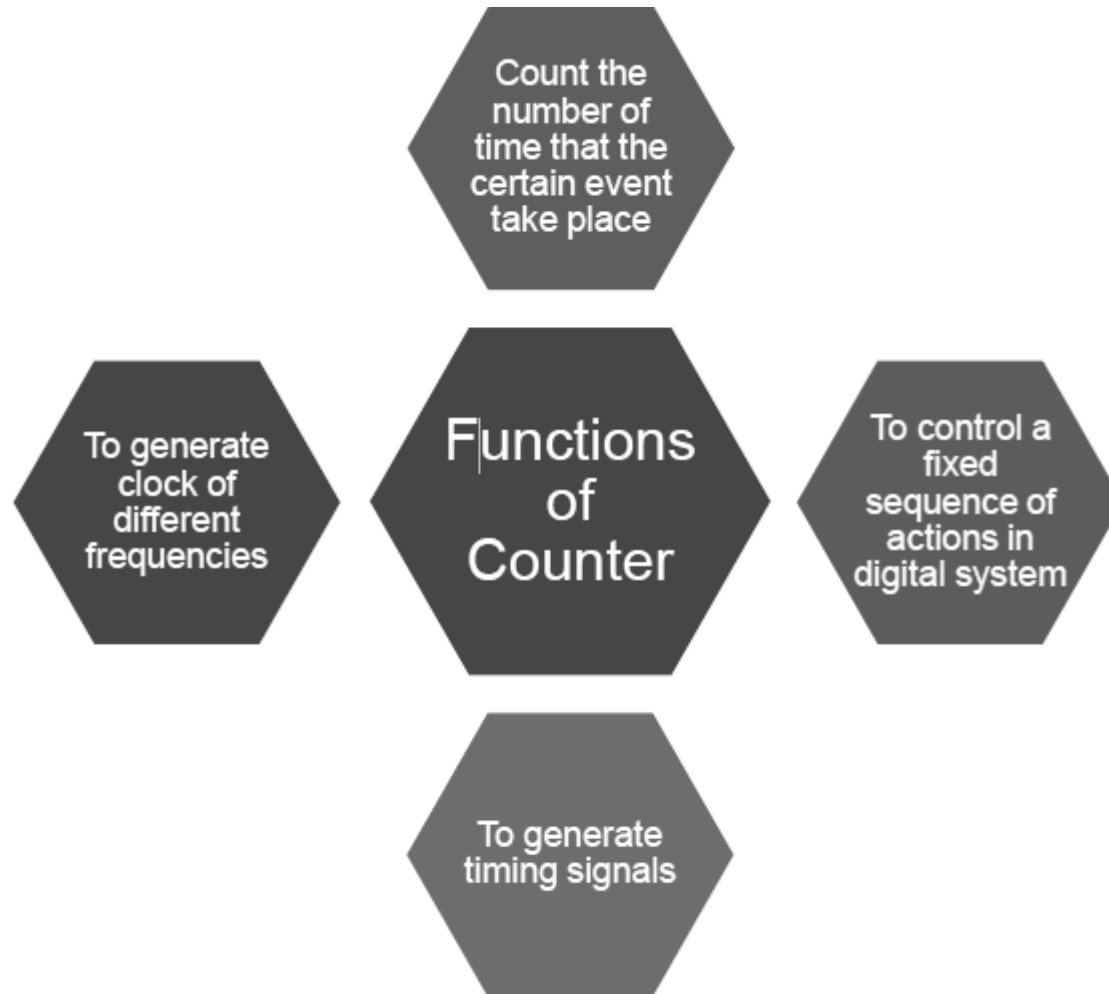
2. Synchronous counter

In the case of synchronous FFs, all the flip flops are triggered simultaneously by an external clock pulse.

counter



Function Of counter



Asynchronous counter

Advantages of Asynchronous counter :

- Due to accumulation of propagation delay, they are used in low speed circuits.
- They are simple to design.
- They are used in mod n counters, and in divide by n counter that divides the input by n (i.e. n is integer).

Disadvantages of Asynchronous counter :

- As the number of flip-flops increases, the propagation delay also increases.
- For high clock frequencies, counting errors may occur because of propagation delay.

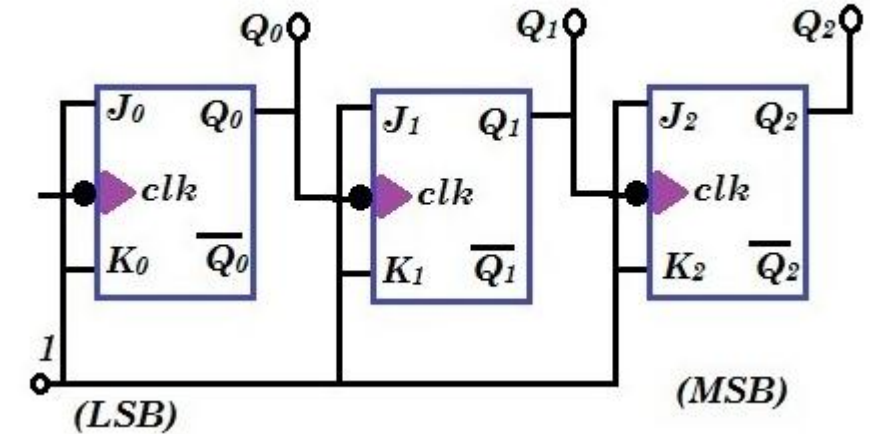
3 bit Asynchronous up counter

A **3-bit asynchronous up counter** is a counter made from 3 **flip-flops** (typically T or JK), where each flip-flop toggles based on the output of the previous one. It counts **upward** ($0 \rightarrow 1 \rightarrow 2 \dots \rightarrow 7$) in binary.

Also called a **ripple counter**, because the clock signal "ripples" from one flip-flop to the next (not all FFs are triggered by the same clock).

Components:

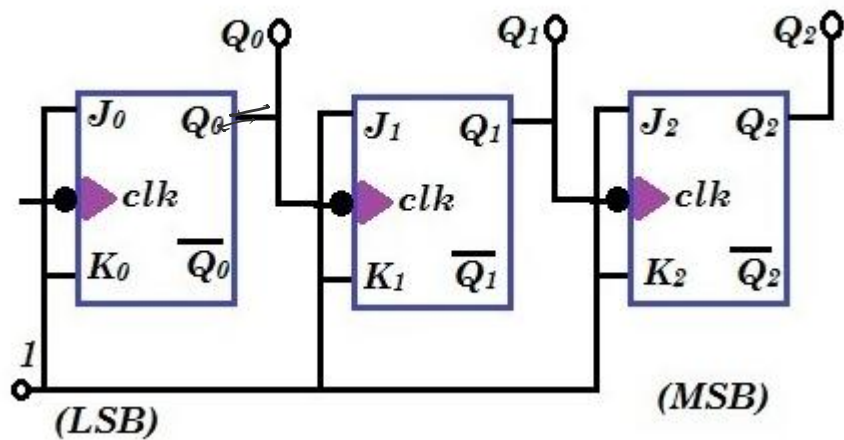
- **3 T flip-flops** (or JK flip-flops configured as T)
- **Clock input** connected **only to the first flip-flop**
- Output of one FF is used as the **clock input for the next**



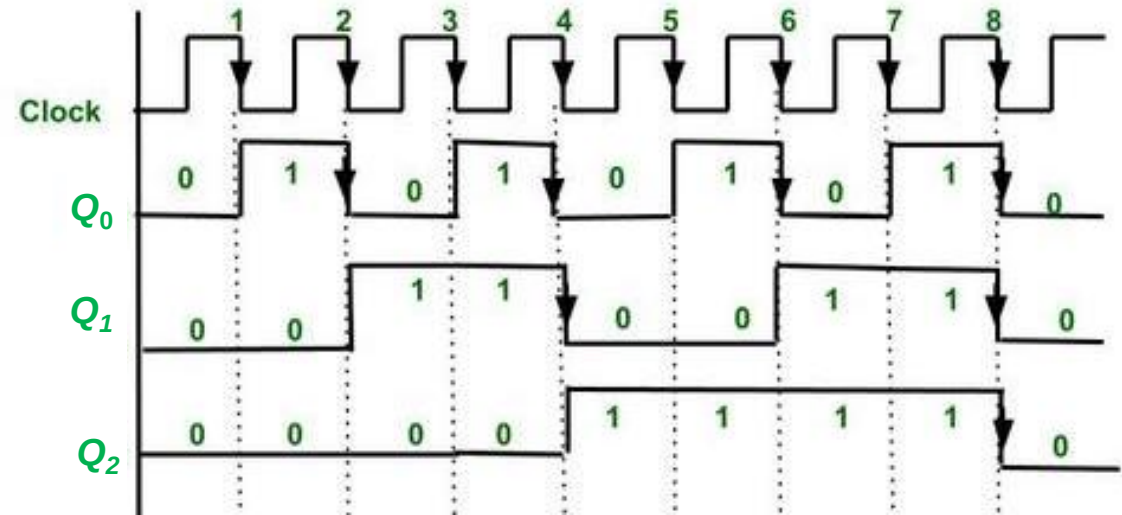
Clocking:

- FF0 toggles on **every** clock pulse
- FF1 toggles on the **falling edge** of FF0's output
- FF2 toggles on the **falling edge** of FF1's output

3 bit Asynchronous up counter



Timing Diagram

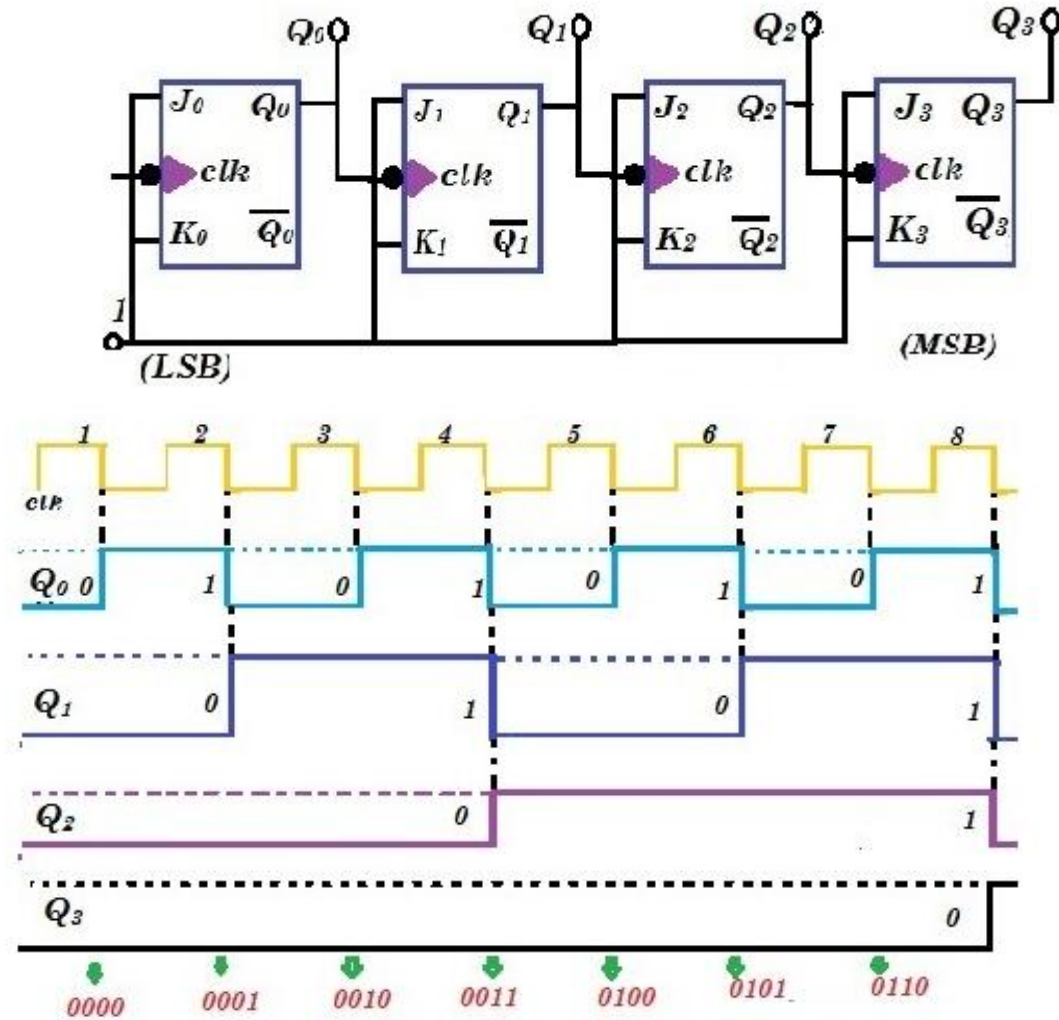


Explanation of Up counter –

- The 1st FF is connected to logic clk. Therefore, it will toggle for every falling edge.
- The 2nd FF input is connected to Q_0 . Therefore it changes its state when $Q_0 = 1$ and there is falling edge of clock.
- Similarly, 3rd FF is connected to Q_1 . Therefore, it changes its state when $Q_1 = 1$ and there is falling edge of clock.
- By this we can generate counting states of Up counter.
- After every 8th falling edge the counter is again reaching to state 0 0 0. Therefore, it is also known as divide by 8 circuit or mod 8 counter.

4 bit Asynchronous up counter

A **4-bit asynchronous up counter** is a digital circuit made of 4 flip-flops connected in series, where each flip-flop counts when the previous one changes from 1 to 0. It counts **from 0 to 15** in binary, increasing by 1 with each clock pulse. It is called *asynchronous* because not all flip-flops get the clock signal at the same time

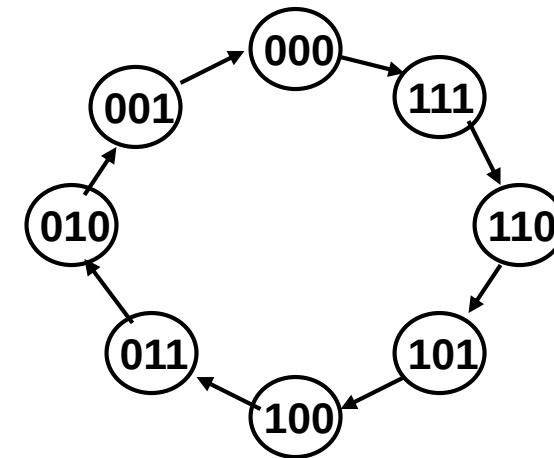
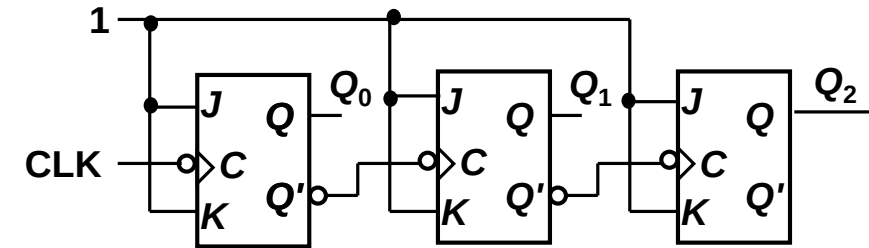


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Asynchronous A 3-bit binary (MOD-8) Down Counters

A **3-bit asynchronous down counter** is a digital circuit that counts **backward** in binary, from **7 (111)** to **0 (000)**. It uses **3 flip-flops**, and each flip-flop toggles when the previous one goes from **0 to 1**. It is called **asynchronous** because the flip-flops are triggered one after another, not all at the same time.

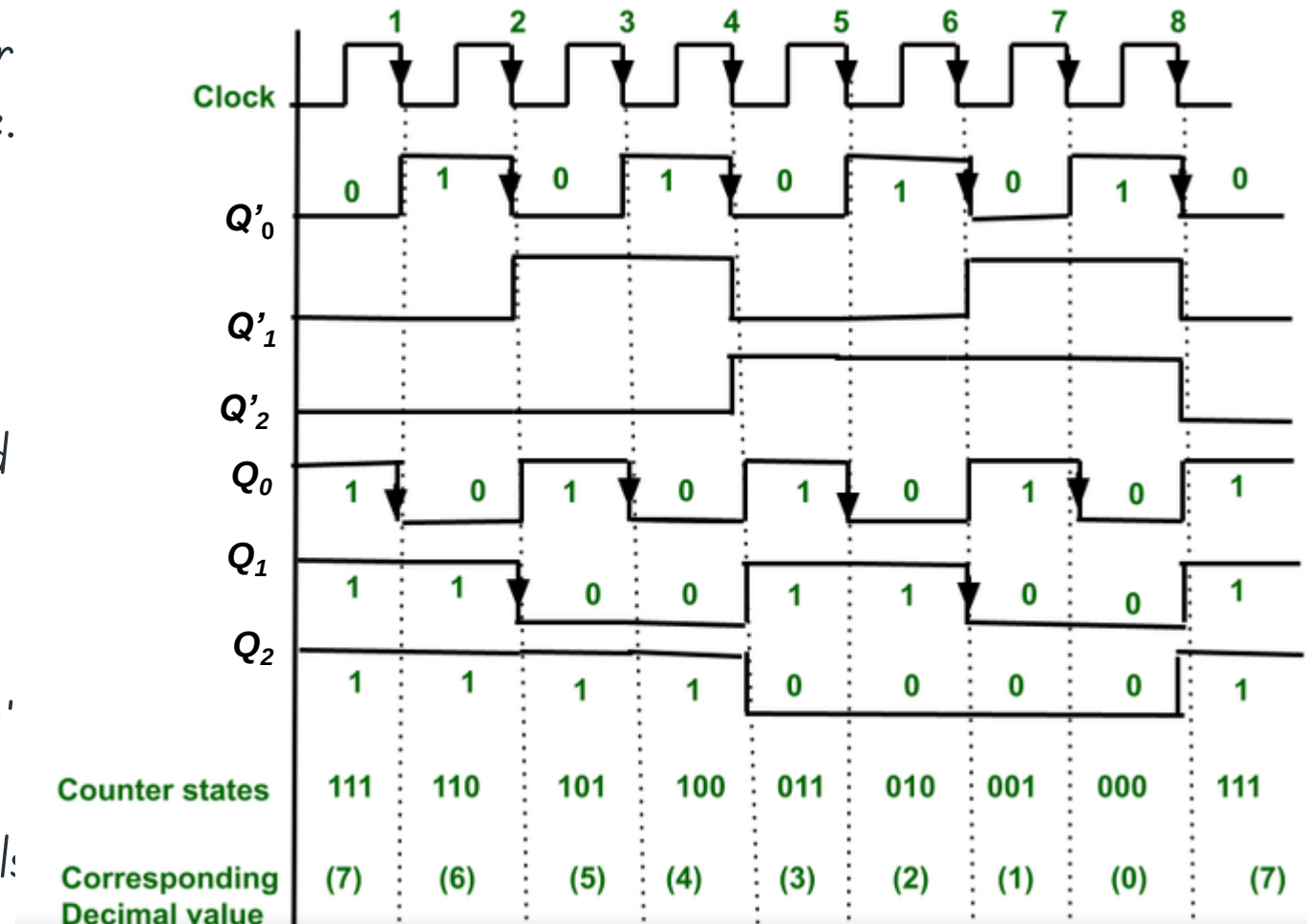
3-bit binary down counter



State diagram

Asynchronous A 3-bit binary (MOD-8) Down Counters

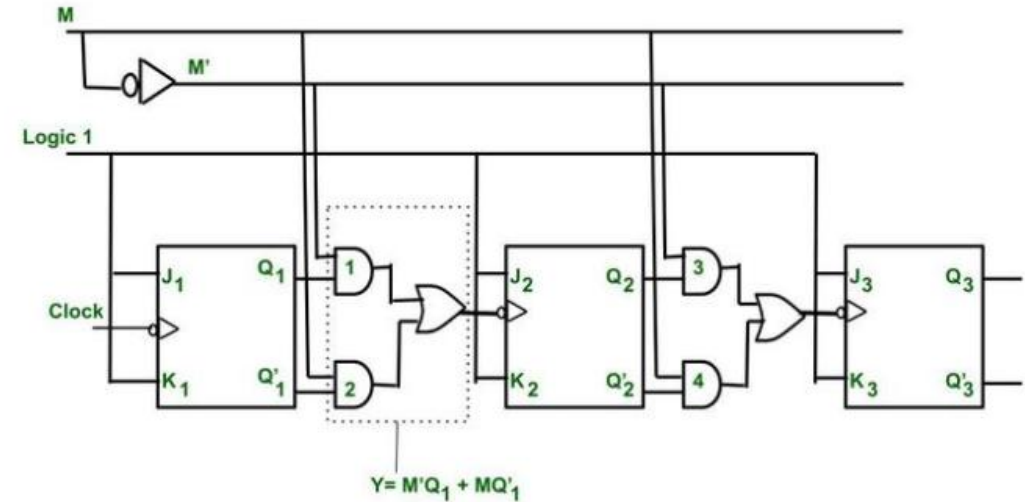
- The -ve edge clock pulse is provided to 1st counter. Therefore, the output state of the first counter (i.e. Q_0) will be toggled at every falling edge of the clock pulse.
- As the Q'_0 is feed as a clock to the second FF, therefore the output state (i.e. Q_1) will be toggled at every falling edge of Q'_0 .
- In the same manner, the Q'_1 acts as clock for the third FF, therefore the output state (Q_2) of the third FF will be toggled for every falling edge of Q'_1 .
- After the 8th falling edge of the external clock pulse the counter is reset to 000



3 bit Asynchronous up/down counter

A **3-bit asynchronous up-down counter with a selector pin** is a counter circuit that can **count up or down** based on the logic level of a **selector pin** (often labeled as **UP/DOWN** or **MODE**). It uses **3 flip-flops** and the selector controls the **counting direction**

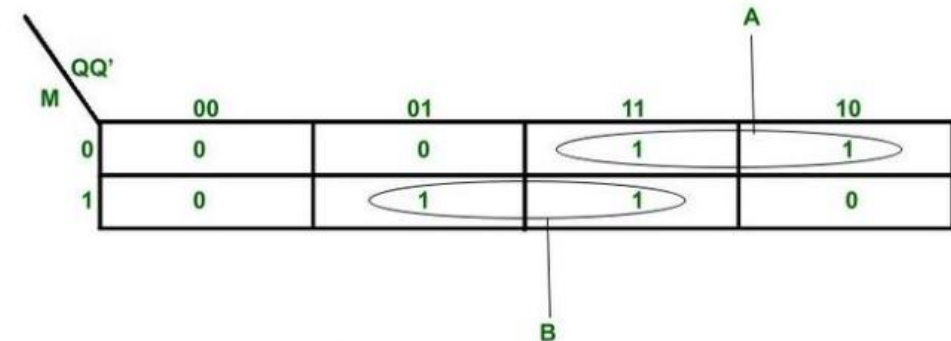
When $M = 0$, then $Y = Q$, therefore it will perform Up counting
 When $M = 1$, then $Y = Q'$ therefore it will perform Down counting.
 $Y = \text{Clock of 2nd/3rd flip flop}$



$Y = Q \text{ when } M=0$
 $Y = Q' \text{ when } M=1$

M	Q	Q'	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Here Y is the output function



$$Y = A + B$$

$$Y = M'Q + MQ'$$

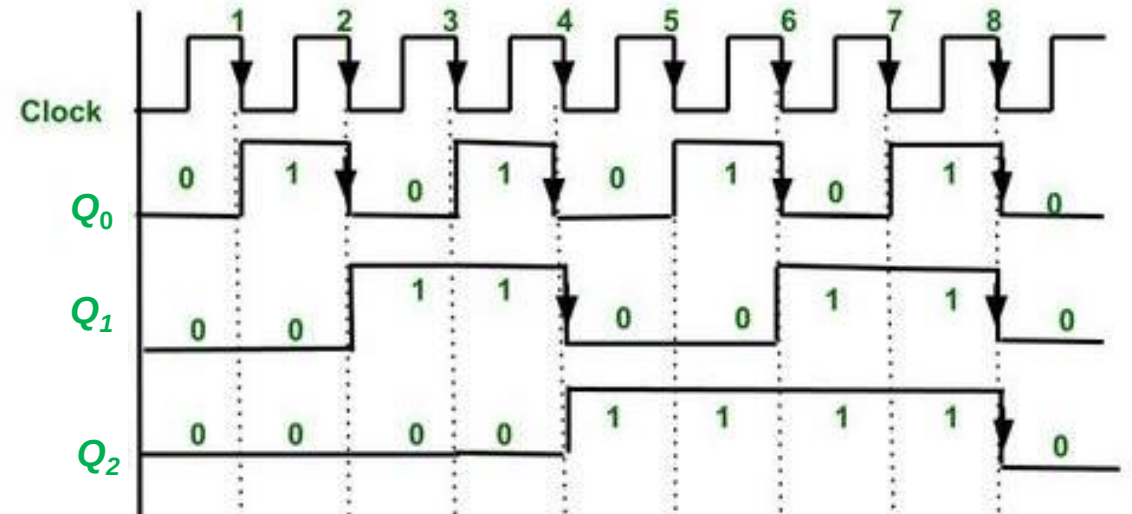
K map for finding Y

Design of 3 bit Asynchronous up/down counter

Case 1 – When $M=0$, then $M'=1$.

Put this in $Y = M'Q + MQ' = Q$ So Q is acting as clock for next FFs.

Therefore, the counter will act as Up counter.



Explanation of Up counter –

When $M = 0$ Up Counter

- The 1st FF is connected to logic 1. Therefore, it will toggle for every falling edge.
- The 2nd FF input is connected to Q_0 . Therefore it changes its state when $Q_0=1$ and there is falling edge of clock.
- Similarly, 3rd FF is connected to Q_1 . Therefore, it changes its state when $Q_1=1$ and there is falling edge of clock.
- By this we can generate counting states of Up counter.
- After every 8th falling edge the counter is again reaching to state 0 0 0.

Therefore, it is also known as divide by 8 circuit or mod 8 counter.

Design of 3 bit Asynchronous up/down counter

Design of 3 bit Asynchronous up/down counter :

Case 2 - When $M=1$, then $M'=0$.

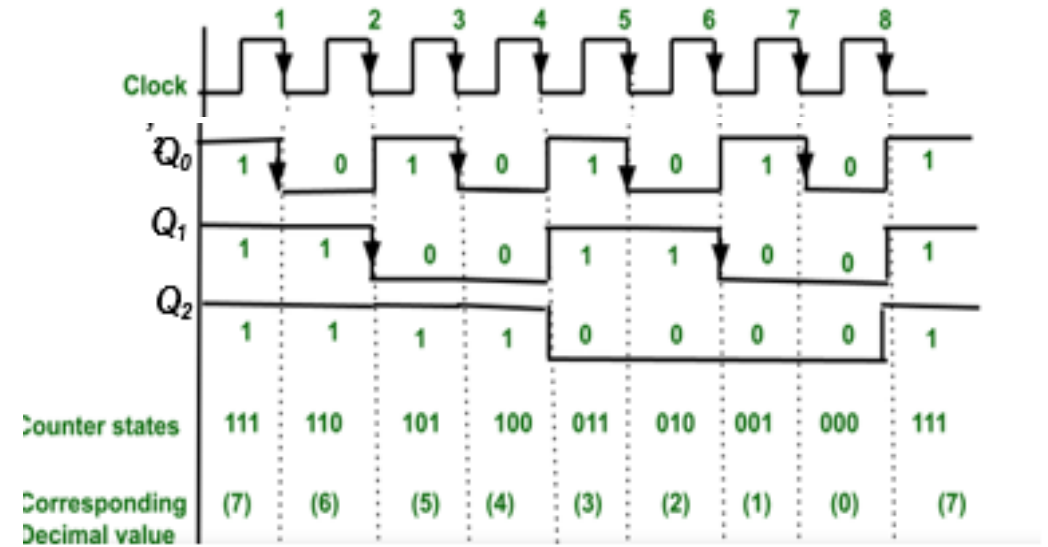
Put this in $Y = M'Q + MQ' = Q'$. So Q' is acting as clock for next FFs.

Therefore, the counter will act as Down counter.

Explanation of Down counter -

- The 1st FF is connected to logic 1. Therefore, it will toggle for every falling edge.
- The 2nd FF input is connected to Q'_0 . Therefore it changes its state when $Q'_0=1$ and there is falling edge of clock.

- Similarly, 3rd FF is connected to Q'_1 . Therefore, it changes its state when $Q'_1=1$ and there is falling edge of clock.
 - By this we can generate counting states of down counter.
 - After every 8th falling edge the counter is again reaching to state 0 0 0.
- Therefore, it is also known as divide by 8 circuit or mod 8 counter.



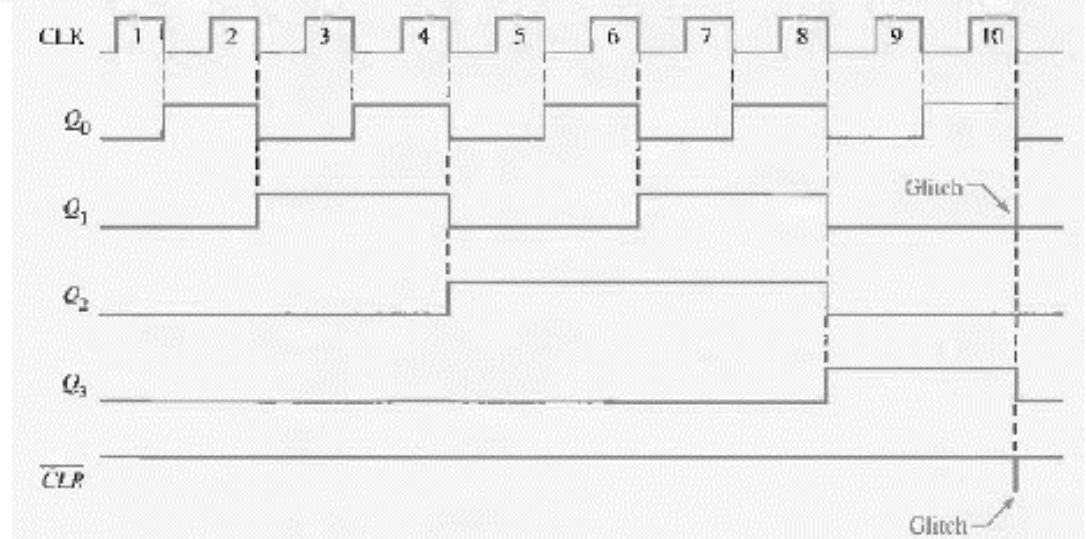
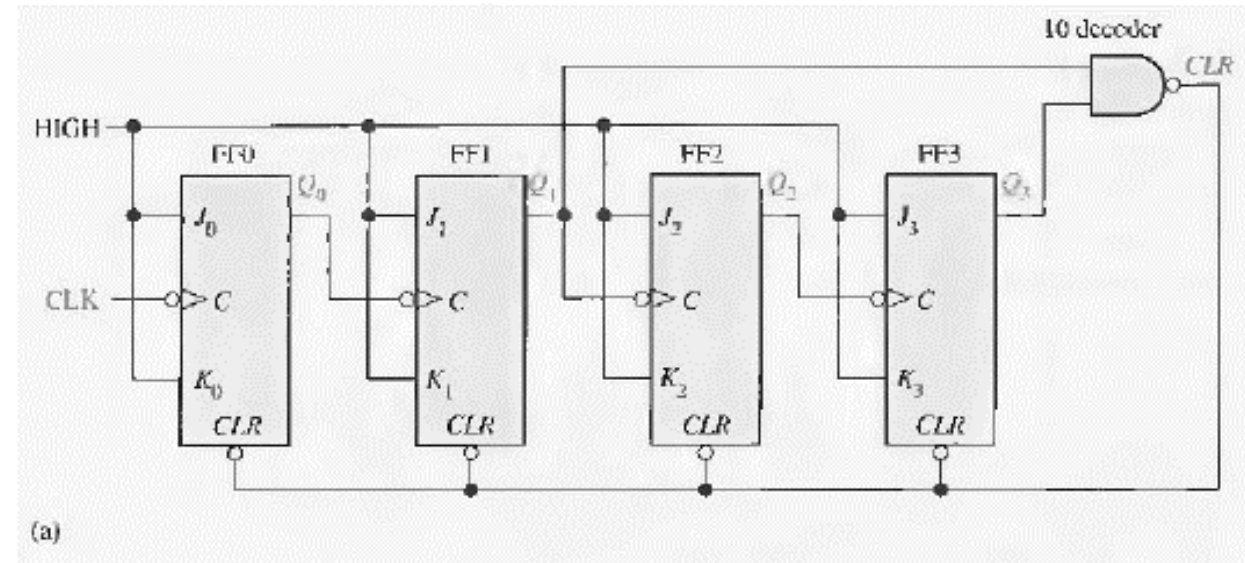
When $M = 1$ Down Counter

Asynchronous Decade Counters

An **asynchronous decade counter** is a **4-bit asynchronous (ripple) counter** that **counts from 0 to 9 (decimal)** in binary form. After counting **9 (1001_2)**, it **resets back to 0 (0000_2)**, making it a **MOD-10** counter

How It Works:

- Uses **4 flip-flops (usually JK or T type)**.
- When the count reaches **1010 (decimal 10)**, it triggers a reset.
- The reset logic is usually done by **NANDing Q_1 and Q_3 ($Q_1 \cdot Q_3 = 1$ at 1010)** → causes **all flip-flops to reset to 0**



Synchronous Counters

A **Synchronous Counter** is a type of digital counter in which **all flip-flops are triggered simultaneously** by a common clock signal.

Unlike asynchronous counters, where the clock ripple causes delays, synchronous counters offer faster and more predictable performance because all flip-flops change state at the same time.

Key Features of Synchronous Counter:

- All flip-flops are driven by the **same clock pulse**.
- **No ripple effect** (unlike asynchronous counters).
- Faster and more suitable for high-speed applications.
- Can be designed to **count up**, **count down**, or both (up/down)

Synchronous Counters

Design Procedure for a Synchronous Counter:

Step 1: Determine the Number of Flip-Flops (n)

Calculate the number of T or J K flip-flops required using the formula:

$2^n \geq N$ Example: For MOD-8 counter $\rightarrow 2^3=8$, so 3 flip-flops are needed.

Step 2: Construct the State Table

Step 3: Derive Boolean Expressions for Flip-Flop Inputs: Use **K-maps** or logic simplification to derive expressions

Step 4: Draw the Logic Circuit

2 Bit Synchronous UP Counters

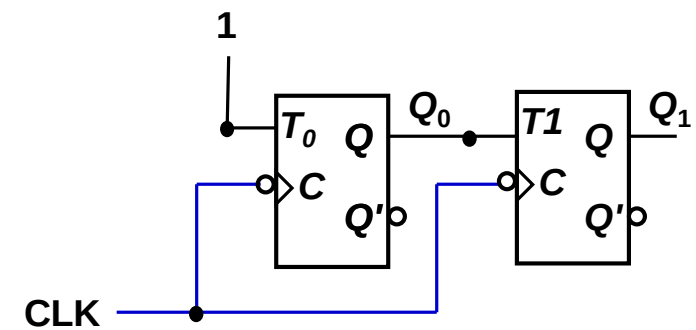
A **2-bit synchronous up counter** is a digital circuit that **counts from 0 to 3** in binary form, increasing by 1 on every clock pulse. It is called **synchronous** because **all flip-flops receive the clock signal at the same time**, so their outputs change **simultaneously**.

Example: 2-bit synchronous binary counter (using T flip-flops, or JK flip-flops with identical J,K inputs).

State table

Present state		Next state		Flip-flop inputs	
Q_1	Q_0	Q_1^+	Q_0^+	T_1	T_0
0	0	0	1	0	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	0	0	1	1

$$\begin{aligned} T_1 &= Q_0 \\ T_0 &= 1 \end{aligned} \quad \text{from K map (do yourself)}$$



Circuit Diagram

3 bit Synchronous (Parallel) Counters

A **3-bit synchronous up counter** is a digital counter that counts from 0 to 7 (i.e., $2^3 = 8$ states) in binary, increasing by 1 on each clock pulse. Since it's *synchronous*, all the flip-flops are triggered **simultaneously** by the same clock signal.

Present state			Next state			Flip-flop inputs		
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

State Table

3 bit Synchronous (Parallel) Counters

Q_1Q_0		00		01		11		10	
		Q_2							
		0	1	0	1	0	1	0	1
0		1	1	1	1	1	1	1	1
1		1	1	1	1	1	1	1	1

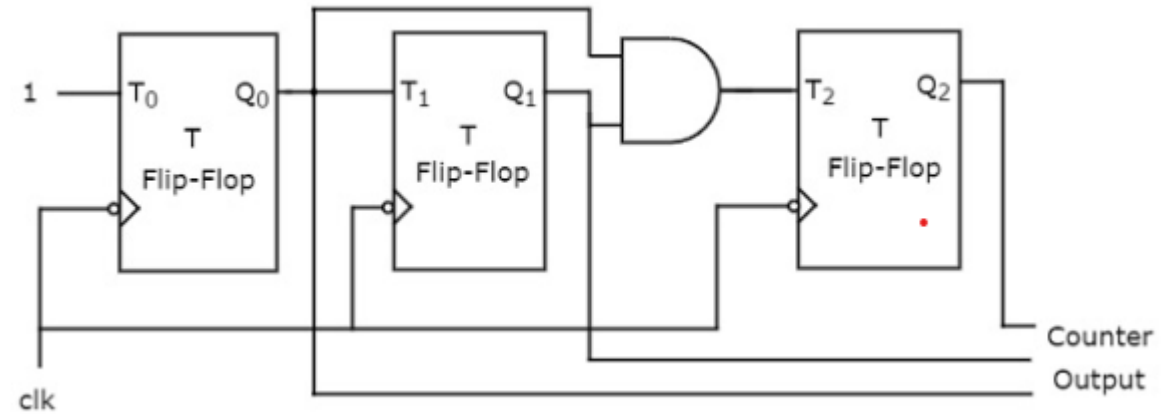
$$T_0 = 1$$

Q_1Q_0		00		01		11		10	
		Q_2							
		0	1	0	1	0	1	0	1
0		0	1	1	0	0	1	0	1
1		0	1	1	0	0	1	0	1

$$T_1 = Q_0$$

Q_1Q_0		00		01		11		10	
		Q_2							
		0	1	0	1	0	1	0	1
0		0	0	1	0	0	0	1	0
1		0	0	1	0	0	0	1	0

$$T_2 = Q_1 \cdot Q_0$$



Circuit Diagram

4 bit Synchronous Decade Counters

A **4-bit synchronous decade counter** counts **from 0 to 9** (0000 to 1001 in binary) and uses **4 flip-flops**. Being **synchronous**, all flip-flops get the clock signal at the same time, so all bits change **together** without ripple delays.

State table

Clock pulse	Present State					Next State					Flip flop Input			
	Q ₃	Q ₂	Q ₁	Q ₀		Q ₃ +	Q ₂ +	Q ₁ +	Q ₀ +		T ₃	T ₂	T ₁	T ₀
Initially	0	0	0	0		0	0	0	1		0	0	0	1
1	0	0	0	1		0	0	1	0		0	0	1	1
2	0	0	1	0		0	0	1	1		0	0	0	1
3	0	0	1	1		0	1	0	0		0	1	1	1
4	0	1	0	0		0	1	0	1		0	0	0	1
5	0	1	0	1		0	1	1	0		0	0	1	1
6	0	1	1	0		0	1	1	1		0	0	0	1
7	0	1	1	1		1	0	0	0		1	1	1	1
8	1	0	0	0		1	0	0	1		0	0	0	1
9	1	0	0	1		0	0	0	0		1	0	0	1

4 bit Synchronous Decade Counters

Q_1Q_0		00	01	11	10
Q_3Q_2	00	0	0	0	0
	01	0	0	1	0
	11	0	1	x	x
	10	x	x	x	x

Q_1Q_0		00	01	11	10
Q_3Q_2	00	0	1	1	0
	01	0	1	1	0
	11	0	0	x	x
	10	x	x	x	x

$T_3 = Q_2 \cdot Q_1 \cdot Q_0 + Q_3 \cdot Q_0$

Q_1Q_0		00	01	11	10
Q_3Q_2	00	0	0	1	0
	01	0	0	1	0
	11	0	0	x	x
	10	x	x	x	x

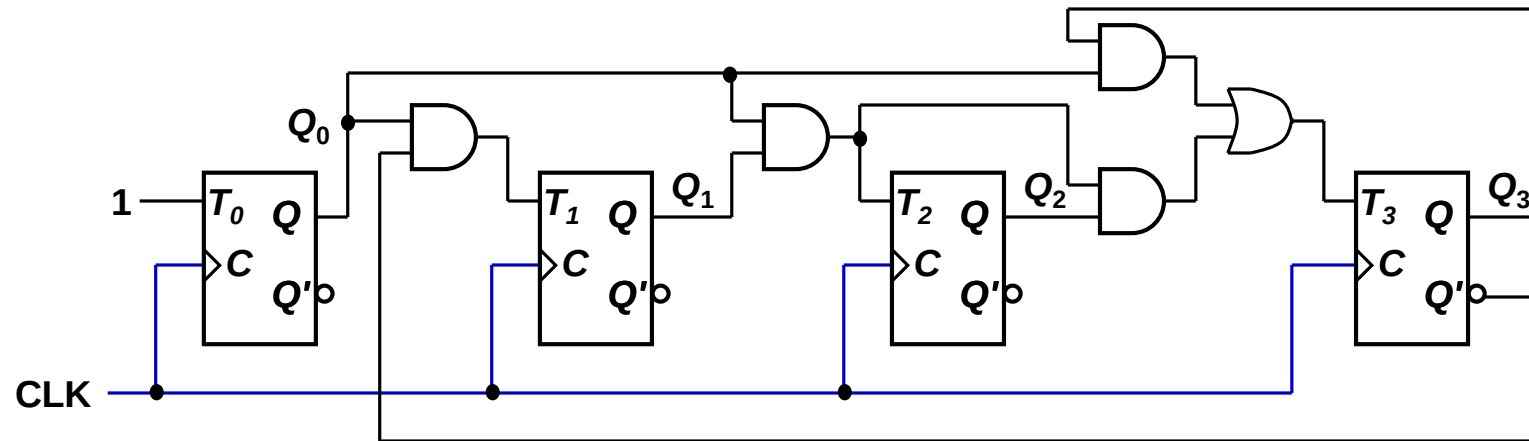
$T_1 = Q'_3 \cdot Q_0$

Q_1Q_0		00	01	11	10
Q_3Q_2	00	1	1	1	1
	01	1	1	1	1
	11	1	1	x	x
	10	x	x	x	x

$$T_2 = Q_1 \cdot Q_0$$

$$T_0 = 1$$

4 bit Synchronous Decade Counters



Circuit Diagram

Arbitrary Sequence Counters Design

Step 1: The number of flip-flops required to design counter can be determined by using the equation: $2^n \geq N$, where n is equal to the number of flip-flops and N is the mod number.

Step 2: Drawing the state diagram.

Step 3: make state table

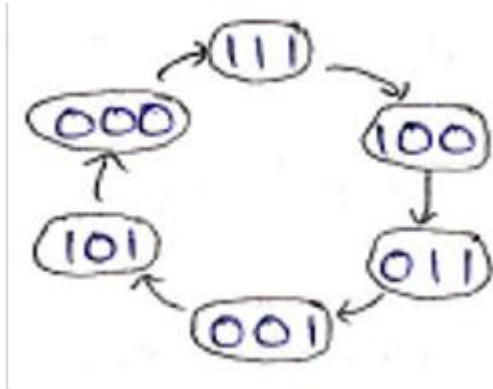
Step 4 Draw circuit

Arbitrary Sequence Counters Design

Design a synchronous counter to count the following sequence 7,4,3,1,5,0,7,4,3,...
Implement the circuit using T-flip-flops.

Step 1: Total number of states is 6. the required number of flip-flops is 3.

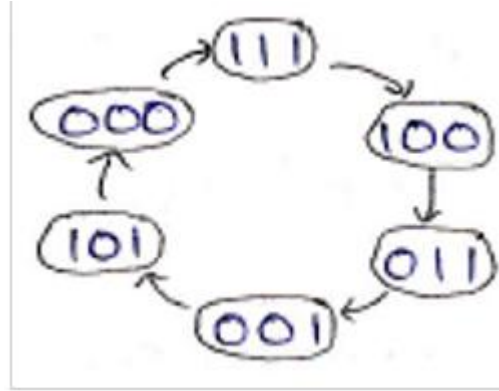
Step 2: State diagram



0 -
1 ->

Arbitrary Sequence Counters Design

Step 2



Solving for $T_1 \rightarrow$

T_1	Q_2	Q_1	Q_0	00	01	11	10
0	1			1		X	
1	1			1		X	

$$T_1 = Q_1 + Q_0'$$

Solving for $T_2 \rightarrow$

T_2	Q_2	Q_1	Q_0	00	01	11	10
0	1	1					X
1	1	1					X

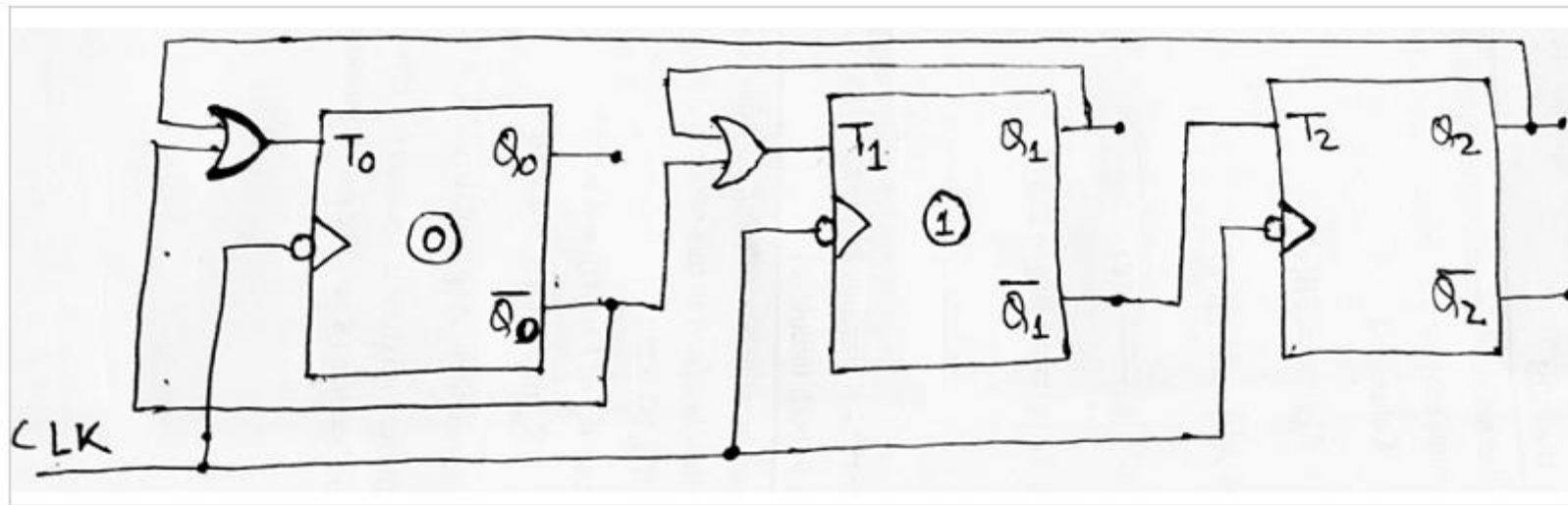
$$T_2 = \overline{Q_1}$$

Solving for $T_0 \rightarrow$

T_0	Q_2	Q_1	Q_0	00	01	11	10
0	1						X
1	1	1	1				X

$$T_0 = Q_2 + \overline{Q_0}$$

Arbitrary Sequence Counters Design



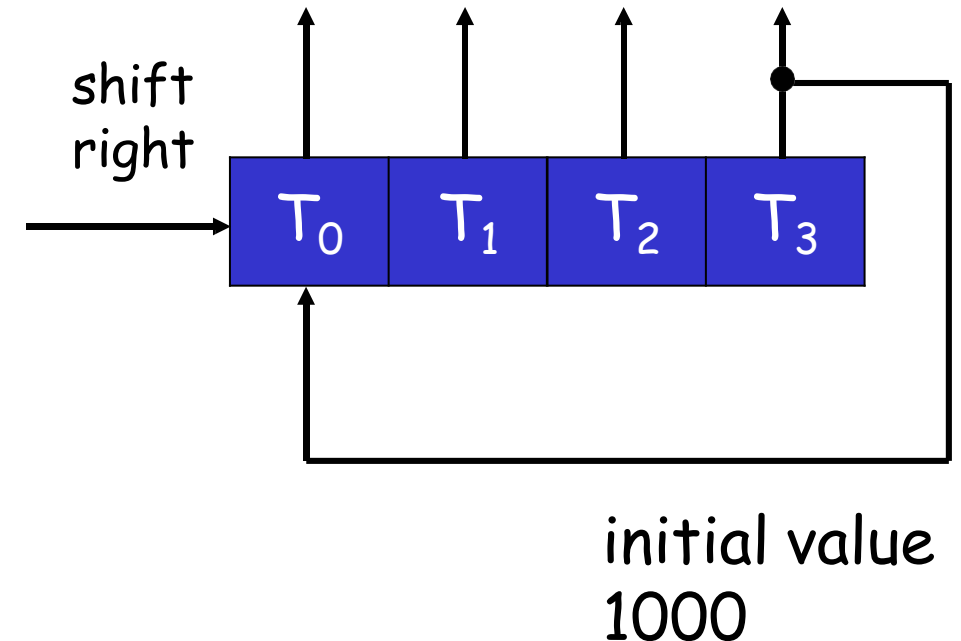
Circuit Diagram

Ring counter

- A **Ring Counter** is a type of **sequential circuit** built using **D flip-flops** connected in a **circular manner**. It works like a shift register, where the output of the last flip-flop is connected back to the input of the first flip-flop

Construction & Working:

- Consists of **n flip-flops**.
- One flip-flop is **preset to '1'**, all others to '0'.
- On each **clock pulse**, the '1' bit **shifts** to the next flip-flop.
- After **n clock cycles**, it returns to the initial state – forming a "ring."



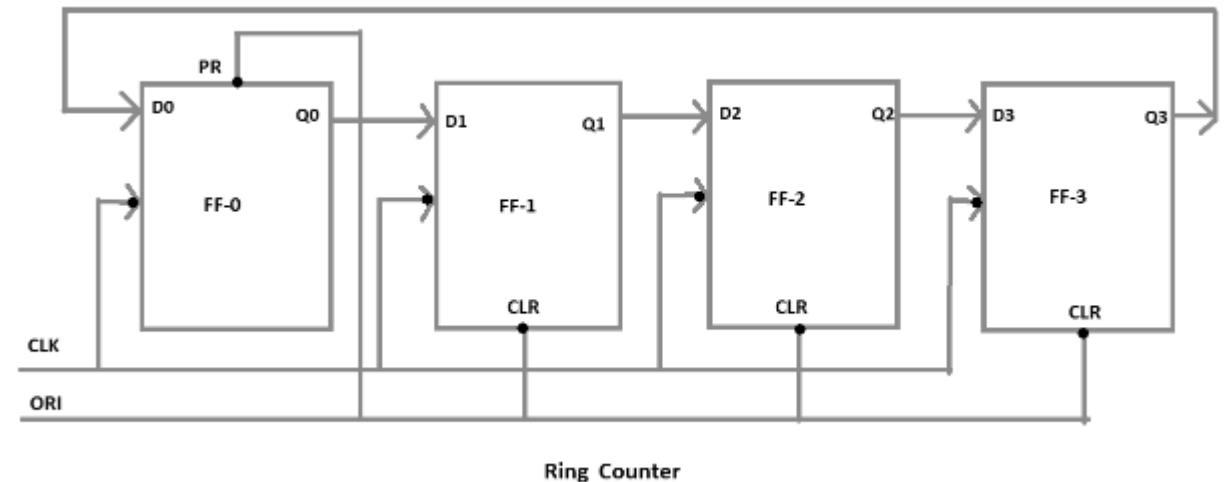
4 bit Ring counter

clock pulse (CLK) is applied to **all the flip-flops** simultaneously. Therefore, it is a Synchronous Counter.

Overriding input (ORI) for each flip-flop. Preset (PR) first FF and Clear (CLR) are used as ORI for other FF.

When PR is 0, then the output is 1. And when CLR is 0, then the output is 0. Both PR and CLR are active low signal that always works in value 0.

ORI is connected to Preset (PR) in FF-0 and it is connected to Clear (CLR) in FF-1, FF-2, and FF-3. Thus, output $Q = 1$ is generated at FF-0, and the rest of the flip-flop generates output $Q = 0$. This output $Q = 1$ at FF-0 is known as Pre-set 1 which is used to form the ring in the Ring Counter.

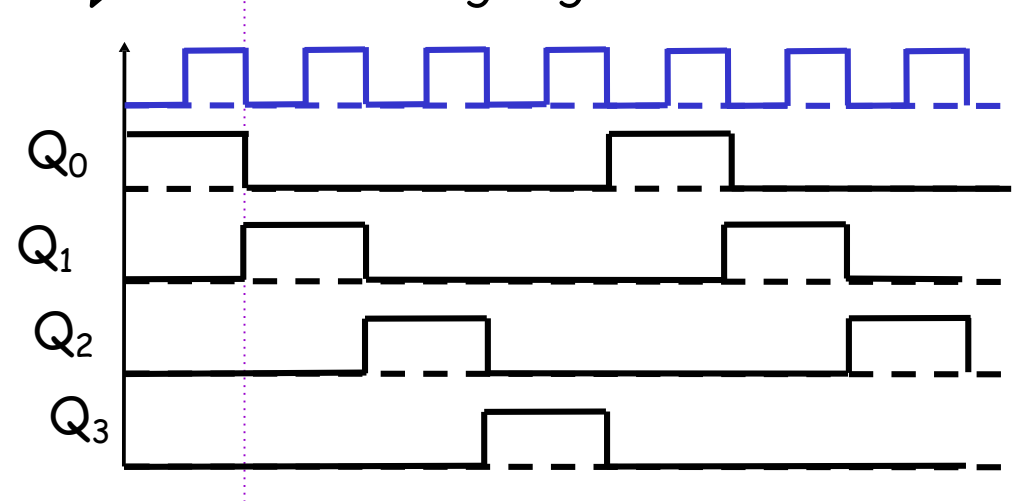


4 bit Ring counter

ORI	CLK	Q0	Q1	Q2	Q3
low	X	1	0	0	0
high	low	0	1	0	0
high	low	0	0	1	0
high	low	0	0	0	1
high	low	1	0	0	0

PRESETED 1

- Sequence of timing signals



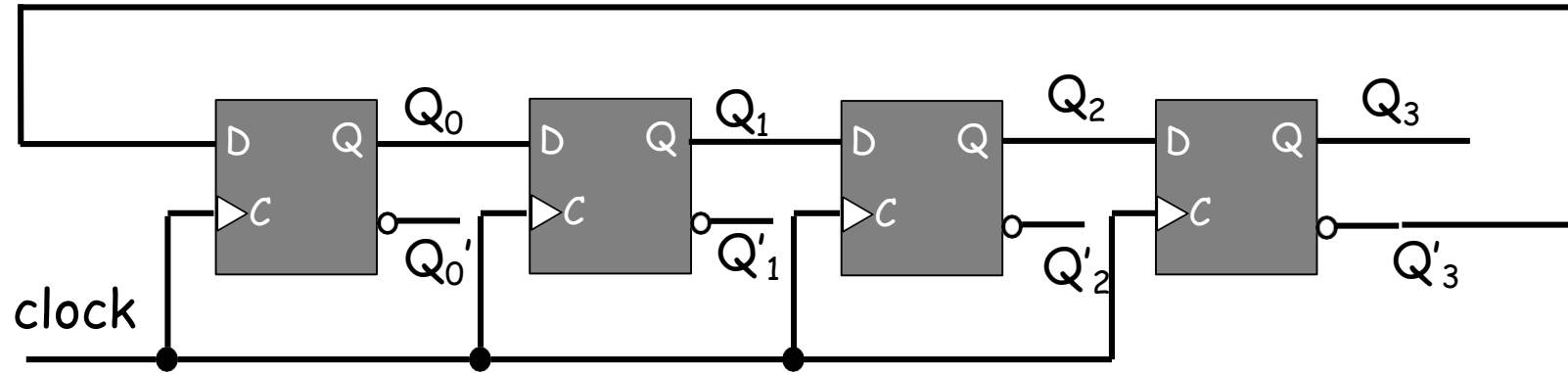
at each clock pulse, the presetted 1 is shifted to the next flip-flop and thus forms a Ring.

From the above table, we can say that there are 4 states in a 4-bit Ring Counter.

Johnson counter

A **Johnson Counter** is a type of modified ring counter where the **inverted output** of the last flip-flop is fed back to the input of the first flip-flop.

This causes a unique sequence of outputs with **twice as many states** as the number of flip-flops used.



Construction & Working:

- Uses **n D flip-flops** connected in series.
- Output of last flip-flop is **inverted and fed back** to the input of the first flip-flop.

Johnson counter

Example with 4 Flip-Flops:

Characteristics:

- Total states = $2n$ (double the flip-flops).
- Only **half the 2^n states** are valid (i.e., not all possible bit combinations are used).
- Efficient use of flip-flops compared to a ring counter

State	Q0	Q1	Q2	Q3
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1
8	0	0	0	0

Ring Vs Johnson counter

Aspect	Ring Counter	Johnson Counter (Twisted Ring)
Definition	A circular shift register with the last output fed directly to the first input.	A shift register where the inverted last output is fed back to the first input.
Feedback Type	Direct ($Q_n \rightarrow D_0$)	Inverted ($Q'_n \rightarrow D_0$)
Number of States	n states (equal to the number of flip-flops)	$2n$ states (double the number of flip-flops)
Flip-Flops Used	n	n
Initial Setup	One flip-flop is set to '1', rest to '0'	All flip-flops reset to '0' (or a known state)
Complexity	Simple	Slightly more complex due to inversion
Unused States	Many unused states (out of 2^n possibilities)	Also has unused states, but utilizes more than Ring Counter
Example (4 FFs)	$1000 \rightarrow 0100 \rightarrow 0010 \rightarrow 0001 \rightarrow 1000$	$0000 \rightarrow 1000 \rightarrow 1100 \rightarrow \dots \rightarrow 0001 \rightarrow 0000$
Applications	LED chasers, light sequence display	Timers, sequence generators, finite state machines